

Assignment 1 - Data structures - C++

100%

30.1.2024

Poengsum frakoblet modus:

100%



Legg til kommentar

Anonym vurdering: **nei**

26.4.2024

▼ Detaljer

Introduction

Note: I recommend repeating the classes lecture from last year before starting this assignment.

This is a fairly big assignment that you might not complete in the recommended time. You can come back to this assignment while working on the next ones.

You're going to display a lot of menus with choices in them. Try to make classes and/or functions to do this so you can reuse code.

Handle errors early. If an error in a function is critical so the function can't continue, just return from it instead of nesting scopes. Use boolean return codes to notify the caller of errors. Typical: True is OK, false is error.

Directory name: ikt103g24v/assignments/solutions/assignment_1_1

1.1

Your task is to write an application that keeps track of:

- Students: id, name, e-mail
- Tests: id, course name, test name, date/time
- Test results: test id, student id, grade

Details:

- IDs should be assigned automatically, starting from 1
- IDs are not reused even if an item is deleted (think database auto increment)
- Any student can take any number of tests
- Students are not always present at tests

Use the ideas we have discussed in class to design classes that keeps track of all this information and a way to store them while the application is running. Follow my user interface to pass the tests:

Please choose:

1. Add student
2. Edit student
3. Remove student

4. Add test
5. Edit test
6. Remove test

7. Add test result
8. Edit test result
9. Remove test result

10. Show all info
11. Exit

The choice numbers must match. The text in the menu does not. Once an operation is completed the application should go back to the menu, ready for another choice.

Each choice has to display information needed to perform that choice. As an example, "Edit student" must show a list of students and let you pick one by number, starting from 1 sorted by id.

Add / edit / remove test result should ask for a student picked by number starting from 1 (sorted by id), and a test picked by number starting from 1, sorted by id, then finally ask for the grade. If the student already has a grade it will be replaced, if he/she doesn't the grade is added.

In all edit / remove menus the last choice means back to the menu (do nothing). Example: There are 2 students. Remove student let's you choose 1 or 2 (for the students), or 3 for back.

Removing a student should also remove all results for that student. The same for other data relationships. The model should never be inconsistent. Adding a test result to a student should not display tests the student has already taken, and so on.

Remember: If code systematically you can test early. Start running the tests once you have some basic functionality in place.

Expected output (all of these must be displayed on choice 10):

```
Student details:
student id = 1, name = Christian Auby, email = christian.auby@uia.no

Test details:
test id = 1, course name = IKT103, test name = Exam, date = 2022.06.12

Test results:
student id = 1, course name = IKT103, test id = 1, test name = Exam, grade = 4
```

Areas of interest:

C++ files and header files. Classes, `std::list`, `std::vector`, `std::map`

Important: The select student and select test menus (used on edit and remove menus) must show a short version of the details above. Otherwise the edit and delete tests will fail because my tests will see the old version of the student/test.

Test locally

see *Local testing with Docker* under Assignments.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

Deliver and test in Bamboo

See *Delivery and Bamboo testing* under Assignments.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.